Highlights

Certified return levels: deciding maximum-likelihood extreme-value fits of significant wave height over a public global hindcast
Extended abstract for the agreement of scope, version 0.1 of 2026-10-01

Carlos Toledo

- Each extreme-value fit is proved the likelihood's one maximum in a recorded box.
- 46,602 fits over a public global hindcast; 5,751 refused with the proof of a limit.
- A ranking by a criterion is a decision only where its enclosures separate.
- A library's default call matches none of 60 certified fits; location fixed, 45.
- A published design table decided row by row against the nearest public cell.

Certified return levels: deciding maximum-likelihood extreme-value fits of significant wave height over a public global hindcast

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Abstract

Problem. A return level of significant wave height is read off a fitted distribution, and a fitted distribution is, in practice, the point at which an optimiser stopped. For two of the six families used in the global study of Reis et al. (2026), the likelihood has no interior maximum on many series, and a general-purpose fitter prints a point on the way; for every family, a free location parameter makes the likelihood unbounded, so the same library returns two 100-year levels for one series. **Method.** The method of Reis et al. (2026) — six families by maximum likelihood on the unfiltered series and on daily, weekly and monthly maxima, four goodness-of-fit criteria, 100- and 1000-year levels — is repeated on its own public hindcast with every step decided in outward-rounded interval arithmetic: a fit is a box proved by the Krawczyk operator to hold exactly one zero of the score, with the Hessian proved negative definite over it; the criteria and the levels are enclosures over the box; a ranking is DECIDED only where its enclosures separate, REFUSED otherwise; the generalized gamma is left out only with a proof that its likelihood peaks at its lognormal limit; a seventh family, the GEV, is certified beside the six. **Results.** Over 2,589 open-sea cells, 46,602 fits were attempted: 40,778 certified, 5,751 refused at the limit with the proof; a threshold in place of the proof changes the selected family at 544 cell-blocks and the verdict of 4 of the method paper's ten regional claims, which are decided as boxes (6 hold, 2 do not, 2 undecided). The same decision applied to other people's numbers: a scientific library's default call is the maximum it reports on none of 60 fits (45 with the location fixed), the benchmark marginals of Haselsteiner et al. (2021) span 5.25–14.53 m for one buoy's hourly 100-year level, and 19 of 20 rows of the design table of Bhaskaran et al. (2023) are off the maximum of the nearest public cell. **Novelty.** Not interval arithmetic, which is standard, but the decision grammar — a verdict with a witness, refusal as a result — and the public record, re-run in a browser by the same bytes.

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Scope proposed for the full manuscript

1. *Data.* The Ifremer WAVEWATCH III hindcast GLOBMULTI_ERA5_GLOBCUR_01 (1993–2024, 3-hourly, CC BY-SA): daily maxima at 3,268 nodes (2,589 open sea, 679 touched by sea ice and excluded), the full 3-hourly series at 13 named nodes, and the three NDBC buoys of the environmental-contour benchmark.
2. *The certificate.* The fit as a box; criteria and levels as enclosures; rankings as decisions; the generalized gamma's lognormal limit and the exponentiated Weibull's Gumbel coordinates treated by proof; the GEV beside the six; the delta-method and extremal-index statistics kept apart and labelled.
3. *Results.* The atlas (Table 1, Fig. 1); the ten regional claims decided; the 3-hourly ledger read against the method paper's three findings; the same library called two ways; the benchmark's marginals and a design table decided.
4. *Discussion.* What a certificate says and does not say; what a standard for marginal fits could fix; cost.

Questions for the co-authors. The journal (Ocean Engineering or Applied Ocean Research); whether the ten claims should be re-boxed by their authors before the table is final; whether the GEV section stays or becomes a note; whether a Python verifier of a downloaded cell certificate should be part of this paper or of a software note.

References

Bhaskaran, S.K., et al., 2023. Comparison of extreme wind and waves using different statistical methods in 40 offshore wind energy lease areas worldwide. Energies 16, 6935.

Haselsteiner, A.F., et al., 2021. A benchmarking exercise for environmental contours. Ocean Engineering 236, 109504.

Reis, R.A.N., Guimarães, P.V., Farina, L., Paul, S., de Paula, T.P., Ribeiro, E.O., 2026. Global assessment of extreme value analysis for significant wave height. Ocean Engineering 359, 125841.

Table 1: The family the Anderson–Darling statistic selects among the six at the 2,589 open-sea cells, with the certificate (left) and with a threshold rule in place of the boundary proof (right).

certificate	daily	weekly	monthly
normal	2	9	52
lognormal	355	437	520
Weibull	9	13	69
exp. Weibull	1,597	1,452	1,115
gen. gamma	495	473	455
Gumbel	111	189	330
refused	20	16	48

threshold rule	daily	weekly	monthly
normal	2	9	52
lognormal	477	520	610
Weibull	9	13	69
exp. Weibull	1,619	1,435	1,041
gen. gamma	353	388	361
Gumbel	116	220	445
refused	13	4	11

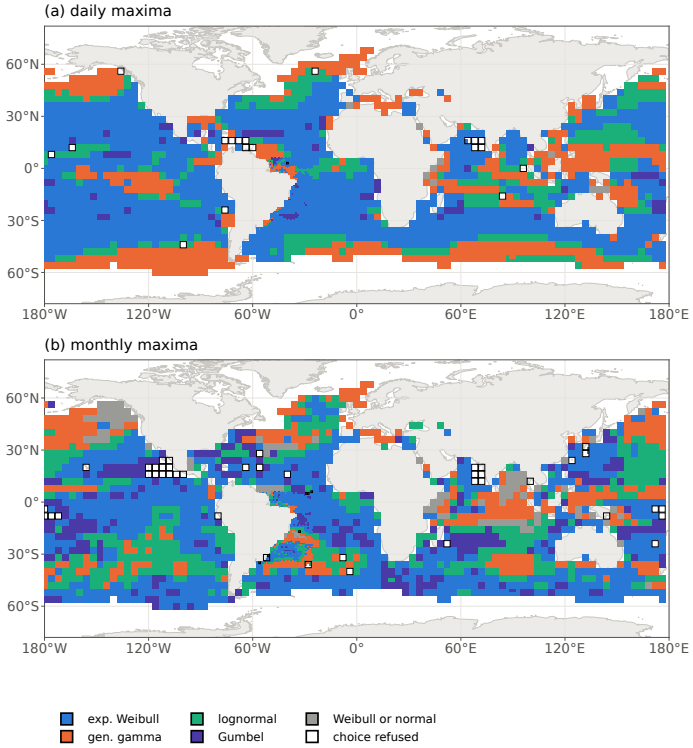


Figure 1: The selected family at every open-sea cell, (a) daily and (b) monthly maxima; hollow squares are refused choices.